RE501 Class 1: Sets and functions

Introduction to Probability

1 Empirical probability

Often in nature, we come across a phenomenon, which gives different results when it repeats - we can not surely predict what will the result be in one occurrence of the phenomenon.

In such cases, we talk about probability of a result to occur.

The simplest way to measure how likely something is: count the ways it can happen, and divide by all the ways things can turn out.

$$\Pr = \frac{\text{number of favorable results}}{\text{total number of possible results}}$$

The answer is always a number between 0 and 1. A probability of 0 means the event never happens; 1 means it always happens.

Quick check. Toss one *fair* coin. There are 2 possible results (Head & Tail) and 1 of them is Head, so Pr(Head) = 1/2.

Clearly, for the above rule to use, we require complete information of what are the possible results.

2 Two words: "Random experiment" and "Outcome"

Random experiment A phenomenon that can repeat, whose result we cannot predict with certainty before it is done, although we know what are the possible results it could give.

Tossing of a coin, rolling a six faced die, and checking the age of the next person you meet are all random experiments.

Outcome A single possible result of a random experiment. Tossing one coin has two outcomes: Head (H) and Tail (T).

With these two words, the ratio definition reads more precisely:

$$\Pr = \frac{\text{number of favourable 'outcomes' of the 'experiment'}}{\text{total number of 'outcomes' of the 'experiment'}}$$

3 Worked example: meeting someone under 30

Question. You walk out of your room. What is the probability that the next person you meet in the office is younger than 30?

- **Random experiment:** meeting the next person in the office.
- **Outcomes:** each person working in the office is one possible outcome.

Suppose the office has 200 people, and 60 of them are younger than 30.

$$\Pr(\text{age} < 30) = \frac{60}{200} = 0.3$$

So roughly 3 out of every 10 people you meet should be under 30.

4 Bias: why the simple ratio is an ideal case

The above probability is estimated assuming **every person in the office is equally likely to be the one you meet**.

Now think of this person:

> A person comes to the office, sits at his desk or lab the entire day, does not go to the canteen or the bank, and after office hours comes out and goes home. He will never meet another person who comes through another gate and sits in another office.

For this person, colleagues in the same room are very likely to be met, and people from the other building have probability close to 0, however many of them there are. Counting heads in the whole office no longer describes his chances.

The ratio 60/200 works only if every outcome has the same chance of occurring - an **ideal** case, which rarely occurs in real life.

When some outcomes are more likely than others for a systematic reason, the experiment is **biased**, and the simple ratio calculation as above gives the wrong answer.

Bias. A systematic tendency of a random experiment to favor some outcomes over others. An experiment with no such tendency is **unbiased**: all its outcomes are equally likely.

5 What kind of thing is an outcome?

Outcomes may or may not be numbers. Rolling a die gives a configuration of die which lands in a way that the top face has definite number of dots (1 to 6), tossing a coin gives a configuration where the top face of the coin is either head or tail, and meeting a person gives a person.

Example: tossing two coins. An outcome is a *configuration*: the way the two coins land. For instance, "HH" is the configuration in which both coins fall with their head on top. Writing the first coin's face first, the four possible outcomes are:

Outcome	Top face of Coin 1	Top face of Coin 2
HH	Head	Head
HT	Head	Tail
TH	Tail	Head
TT	Tail	Tail

We collect all of them together and denote as $\{HH, HT, TH, TT\}$. To talk precisely about such collections of outcomes, and about events as groups of outcomes, we need a new language. That is the subject of the next chapter: **set theory**.

Chapter 1: Set theory

6 What is a set?

Set. A set is a collection of well-defined objects.

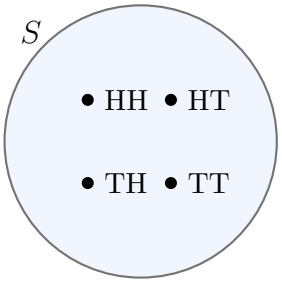
"Well-defined" means that for any object we can say clearly whether it belongs to the collection or not.

- "The faces of a coin" is a set: $\{H, T\}$.
- "The tall students in this class" is **not** a set, because "tall" is not well defined.

The objects in a set are its **elements**. We write $x \in A$ to mean that "x is an element of A" and $x \notin A$ to mean "x is not an element of A". Sets are usually named with capital letters and elements with small letters, unless otherwise specified.

Venn diagram

A Venn diagram draws a set as a closed region, usually a circle. Each element is shown as a point inside the region; anything outside the region is not in the set.



The same set in roster form
$S = \{HH, HT, TH, TT\}$

Roster form

The way we wrote the outcomes of two coin tosses earlier, $\{HH, HT, TH, TT\}$, is called the **roster form** of a set: list every element, separated by commas, inside curly braces. The order of the elements does not matter, and each element is listed once.

7 Finite, infinite sets & cardinality

Finite set A set with finite number of elements.
Examples: $\{H, T\}$; the outcomes of two coin tosses; the people in your office are all finite sets.

Infinite set. A set that has infinite number of elements in it.
Examples: the natural numbers $\{1, 2, 3, \dots\}$; all real numbers between 0 and 1; the number of tosses needed to see the first Head (it could be $1, 2, 3, \dots$) are all infinite sets.

Cardinality. For a finite set A, the cardinality is the number of elements in A, written $n(A)$

$$S = \{HH, HT, TH, TT\} \implies n(S) = 4$$

Note that the ratio definition of probability from Section 1 is a ratio of cardinalities: $\Pr(E) = n(E)/n(S)$, where S is the set of all outcomes. This only makes sense when S is finite.

8 Null set, subset, superset, power set

Null set

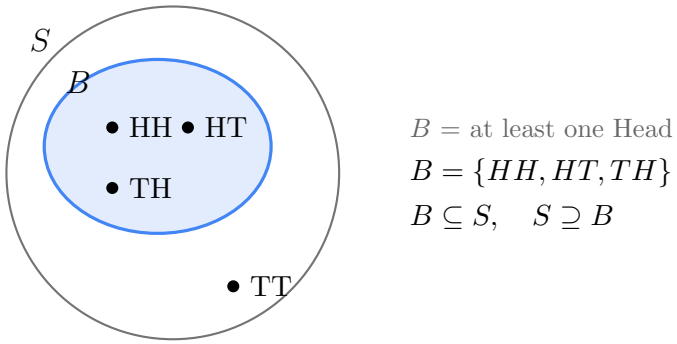
The **null set** (empty set) is the set with no elements, written $\emptyset$ or $\{\,\}$. Its cardinality is $n(\emptyset) = 0$. Example: the set of outcomes of two coin tosses that show three Heads.

Subset and superset

A is a **subset** of B, written $A \subseteq B$, if every element of A is also an element of B. Equivalently, B is a **superset** of A, written $B \supseteq A$.

$$A \subseteq B \iff (x \in A \implies x \in B)$$

In a Venn diagram, a subset is a region drawn entirely inside the larger region.



Useful facts:

- Every set is a subset of itself: $A \subseteq A$.
- The null set is a subset of every set: $\emptyset \subseteq A$.
- If $A \subseteq B$ and $A \neq B$, A is a **proper subset** of B, written $A \subset B$.

An event in probability is a subset of the set of outcomes. "At least one Head" is the subset B above.

Power set

The **power set** of A is the set of all subsets of A. It is denoted 2^A (also $\mathcal{P}(A)$).

$$2^A = \{\, X \mid X \subseteq A \,\}$$

It is important to remember that A is an element of 2^A. This we will emphasize using the notation:

$$\begin{array}{c} 2^A \\ \downarrow \\ A \end{array}$$

Example. Let $A = \{a, b, c\}$. Its subsets are:

Size of subset	Subsets	Count
0	$\emptyset$	1
1	$\{a\}, \{b\}, \{c\}$	3
2	$\{a, b\}, \{a, c\}, \{b, c\}$	3
3	$\{a, b, c\}$	1

$$2^A = \{\, \emptyset,\ \{a\},\ \{b\},\ \{c\},\ \{a, b\},\ \{a, c\},\ \{b, c\},\ \{a, b, c\}\,\}$$

Why there is a 2 in symbol of power set? If we take a finite set A of size n, i.e. $n(A) = n$ then we can build its power set as following-

- choosing no elements to build the subset: $\binom{n}{0}$ such subsets
- choosing 1 element to build the subset : $\binom{n}{1}$ such subsets
- choosing 2 elements to build the subset : $\binom{n}{2}$ such subsets
  ...
- choosing n elements to build the subset: $\binom{n}{n}$ such subsets

So the number of elements in the power set will be

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n$$

Thus a finite set with n elements has 2^n elements in its power set. Perhaps this is the reason why power set is shown as 2^A.

The power set is not limited to finite sets. The definition "the set of all subsets of A" is defined for any set. For example, the power set of the natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ contains $\emptyset$, $\{1\}$, $\{2, 4, 6, \dots\}$ (the even numbers), $\{1, 2, 3\}$ and $\mathbb{N}$ itself, among infinitely many others. However, since the set is not finite, the cardinality rule of power set $n(2^A) = 2^n$ no longer applies here.

In the above, we used binomial expansion:

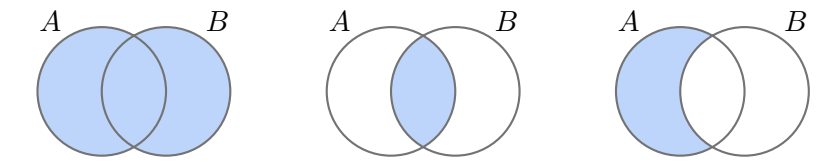
$$(1 + x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$$

Use $x = 1$ to get the required identity.

9 Set operations: union, intersection, difference

Each operation builds a new set from two sets A and B. We describe each one in **set-builder notation**, which defines a set by the property its elements satisfy instead of listing them:

$$\{\, x \mid \text{property of } x \,\} \quad \text{read: "the set of all } x \text{ such that } \dots\text{"}$$



$A \cup B$
in A or in B (or both)

$A \cap B$
in A and in B

$A \setminus B$
in A but not in B

Symbol	Meaning
$a \in A$	element a is in set A
$a \notin A$	element a is not in set A
$A \subseteq B$	set A is a subset of set B
$B \supseteq A$	set B is a superset of set A
2^A	power set of A
$\vee$	or
$\wedge$	and
$\exists$	there exists
$\nexists$	doesn't exist
$\forall$	for all
$\mid$	such that
$\implies$	implies
$\iff$	if and only if (also written as iff)
$\cup$	union
$\cap$	intersection
$A \setminus B$	A minus B
A^C	Complement of set A

Union

The union $A \cup B$ contains every element that is in A, or in B, or in both.

$$x \in A \cup B \iff x \in A \vee x \in B$$

$$A \cup B = \{\, x \mid x \in A \vee x \in B \,\}$$

Intersection

The intersection $A \cap B$ contains only the elements that are in both A and B.

$$x \in A \cap B \iff x \in A \wedge x \in B$$

$$A \cap B = \{\, x \mid x \in A \wedge x \in B \,\}$$

Difference

The difference $A \setminus B$ (also written $A - B$) contains the elements of A that are not in B.

$$x \in A \setminus B \iff x \in A \wedge x \notin B$$

$$A \setminus B = \{\, x \mid x \in A \wedge x \notin B \,\}$$

In general, $A \setminus B \neq B \setminus A$: the order matters.

Example with two coin tosses

Let $S = \{HH, HT, TH, TT\}$, $A =$ "first coin shows Head" $= \{HH, HT\}$, and $B =$ "second coin shows Head" $= \{HH, TH\}$.

Operation	Result	In words
$A \cup B$	$\{HH, HT, TH\}$	at least one Head
$A \cap B$	$\{HH\}$	both Heads
$A \setminus B$	$\{HT\}$	only the first coin shows Head
$B \setminus A$	$\{TH\}$	only the second coin shows Head

10 Universal set, complement, and mutually exclusive sets

Universal set and complement

The **universal set** U is the set of all objects under discussion. In probability, U is the set of all outcomes S; for two coin tosses, $U = \{HH, HT, TH, TT\}$.

The **complement** of A, written A^c (also A' or $\bar{A}$), is the set of all elements of U that are not in A.

$$A^c = \{\, x \mid x \in U \wedge x \notin A \,\} = U \setminus A$$

Example. With $A =$ "first coin shows Head" $= \{HH, HT\}$, the complement is $A^c = \{TH, TT\}$, "first coin shows Tail."

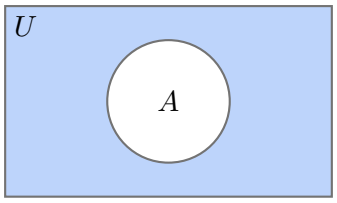
Two facts follow at once: $A \cup A^c = U$ and $A \cap A^c = \emptyset$.

Mutually exclusive sets or disjoint

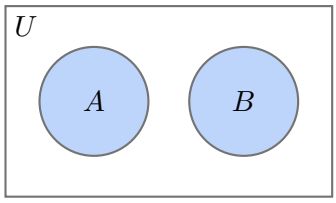
A and B are **mutually exclusive** (disjoint) if they have no element in common.

$$A \cap B = \emptyset$$

Example: "Both Heads" $= \{HH\}$ and "Both Tails" $= \{TT\}$ are mutually exclusive: the two events cannot happen in the same toss. A set and its complement are always mutually exclusive.



A^c (shaded)
in U but not in A



$A \cap B = \emptyset$
mutually exclusive: no common element

11 Counting the elements of a union

Mutually exclusive finite sets

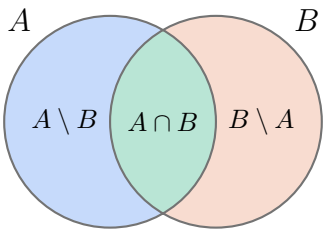
If A and B are finite and mutually exclusive, no element is counted twice, so the counts simply add:

$$A \cap B = \emptyset \implies n(A \cup B) = n(A) + n(B)$$

Example. $n(\{HH\} \cup \{TT\}) = n(\{HH, TT\}) = 2 = 1 + 1$.

Overlapping sets

If A and B overlap, $n(A) + n(B)$ counts the common elements twice. The fix is to break $A \cup B$ into pieces that do **not** overlap and then use the rule above.



$$n(A \cup B) = n(A \setminus B) + n(A \cap B) + n(B \setminus A)$$

Step 1. Split the union into three mutually exclusive pieces:

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$$

$$n(A \cup B) = n(A \setminus B) + n(A \cap B) + n(B \setminus A)$$

Step 2. Split A and B the same way. $A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$, each a disjoint union:

$$n(A \setminus B) = n(A) - n(A \cap B), \qquad n(B \setminus A) = n(B) - n(A \cap B)$$

Step 3. Substitute into Step 1:

$$n(A \cup B) = \big(n(A) - n(A \cap B)\big) + n(A \cap B) + \big(n(B) - n(A \cap B)\big)$$

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

When A and B are mutually exclusive, $n(A \cap B) = 0$ and we get back the first rule.

Example. $A = \{HH, HT\}$, $B = \{HH, TH\}$, $A \cap B = \{HH\}$. Then $n(A \cup B) = 2 + 2 - 1 = 3$, which matches $A \cup B = \{HH, HT, TH\}$.

12 Union as addition, intersection as multiplication

There exists a useful analogy between set operations and ordinary algebra:

Algebra	Set theory
addition, $a + b$	union, $A \cup B$
multiplication, $a \times b$	intersection, $A \cap B$

Using the analogy: the distributive law

In algebra, multiplication distributes over addition:

$$a \times (b + c) = (a \times b) + (a \times c)$$

Replace $\times$ by $\cap$ and $+$ by $\cup$, and we get the corresponding law for sets:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

The analogy is a memory aid, not an exact match: for example $A \cup A = A$, whereas $a + a = 2a$. Always check a guessed law with Venn diagram.

13 De Morgan's laws

Taking the complement turns a union into an intersection, and an intersection into a union:

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

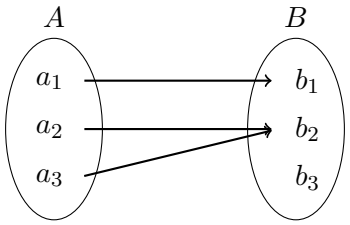
Chapter 2: Function

14 What is a Function?

A function f from a set A to a set B (denoted as $f : A \to B$) is a relation that assigns to each element $a \in A$ exactly one element $b \in B$. We write this as $f(a) = b$ or $a \mapsto b$, where $a \in A$ and $b \in B$.

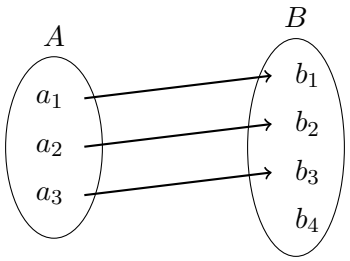
Core criteria of a function:

- **Total Coverage:** It must cover all elements of set A. Every $a \in A$ must have a mapped image in B.
- **Single Value Assignment:** One-to-many mappings are not possible. An element $a \in A$ cannot map to more than one element in B.



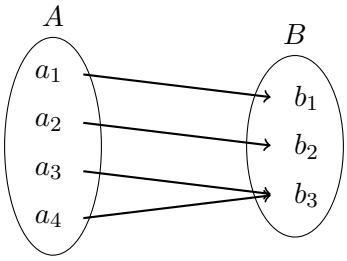
15 One-to-One (Injective) Function

A function $f : A \to B$ is **injective** if distinct elements in A map to distinct elements in B ($f(a_1) = f(a_2) \implies a_1 = a_2$). Below is an example of an injective function where not all elements of B are covered:



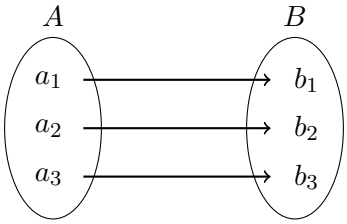
16 Surjective (Onto) Function

A function $f : A \to B$ is **surjective** if every element in B is covered by at least one element in A. Below is a surjective function that is **not injective** (many-to-one):



17 Bijective Function and Invertibility

A function is **bijective** if it is both **injective** and **surjective**, forming a perfect one-to-one correspondence.



Argument for Invertibility: An inverse function $f^{-1} : B \to A$ reverses the mapping direction. For f^{-1} to be a valid function, it must satisfy two conditions:

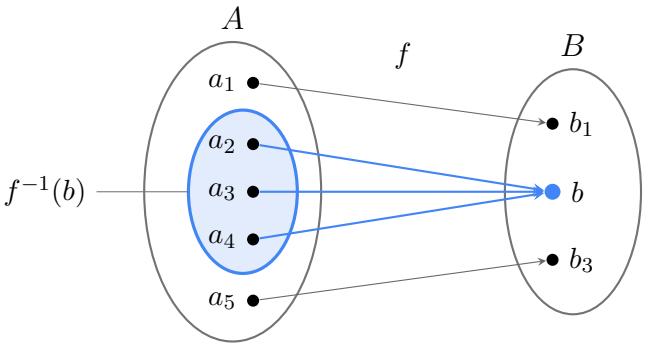
1. **Total Coverage of the new domain (B):** If f is not surjective, there is an element in B with no arrow pointing to it. Reversing the arrow leaves this element unmapped in f^{-1}, violating total coverage.
2. **No One-to-Many Mappings:** If f is not injective (i.e., many-to-one), multiple elements in A point to the same element in B. Reversing the arrows forces that element in B to map to multiple elements in A, creating an invalid one-to-many mapping.

Hence, **only bijective functions** have a well-defined inverse function.

18 Preimage

Consider a surjective but non injective function $f : A \to B$. Since surjective, every element of B is mapped to. Now consider one such element $b \in B$. We ask, what are the elements in A that mapped to b? These elements form a set, called preimage of b, denoted by $f^{-1}(b \in B)$-

$$f^{-1}(b \in B) = \{x \in A \mid f(x) = b\} \subseteq A$$



$$f^{-1}(b) = \{a_2, a_3, a_4\} \subseteq A$$